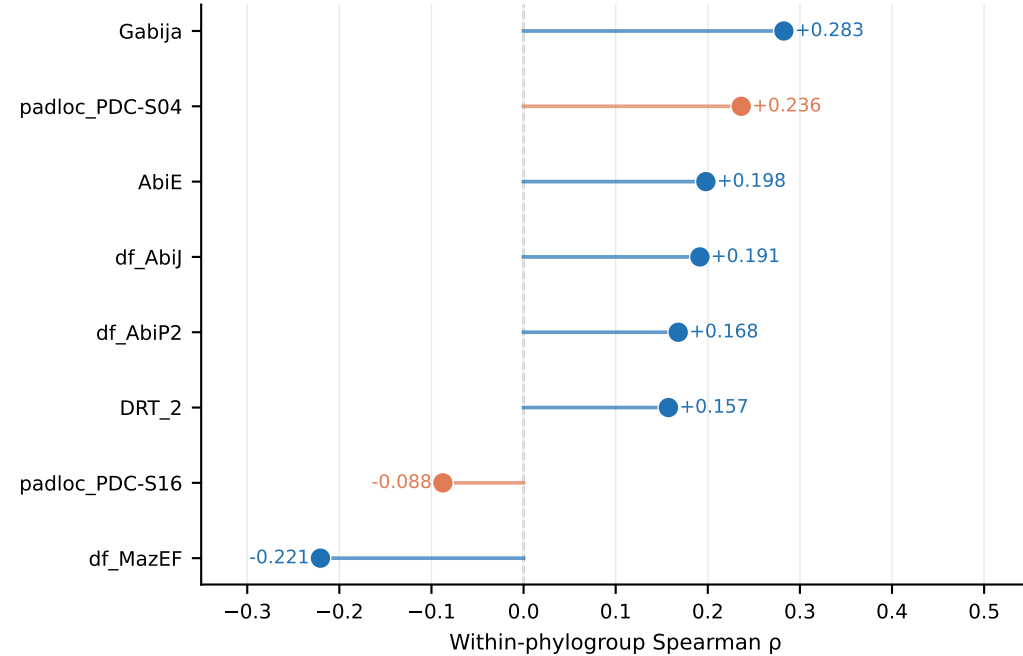
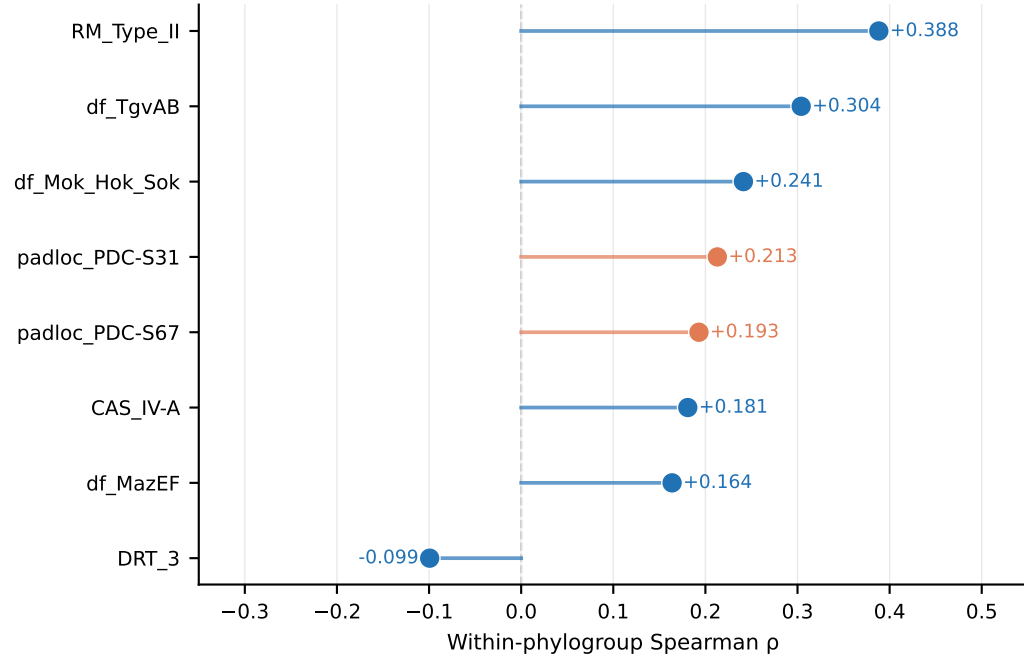


(B) Top within-phylogroup-robust ARG-burden drivers, Q2-significant species
Spearman ρ corrected for phylogroup structure; $p_{\text{within_pg}} < 0.05$

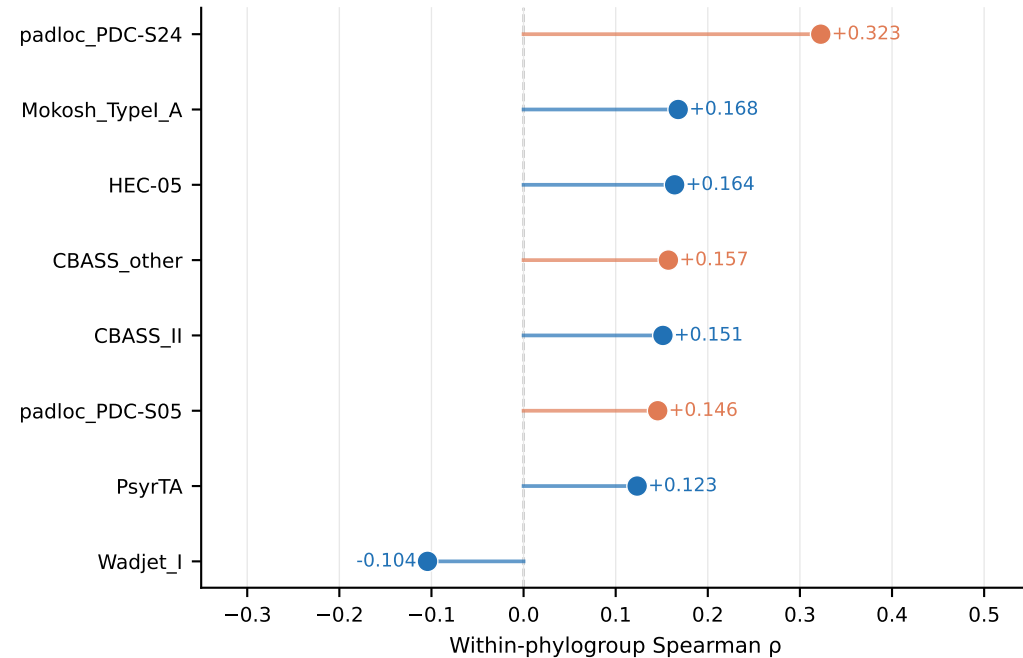
E. faecium
AUROC = 0.824 (10 robust drivers; showing top 8)



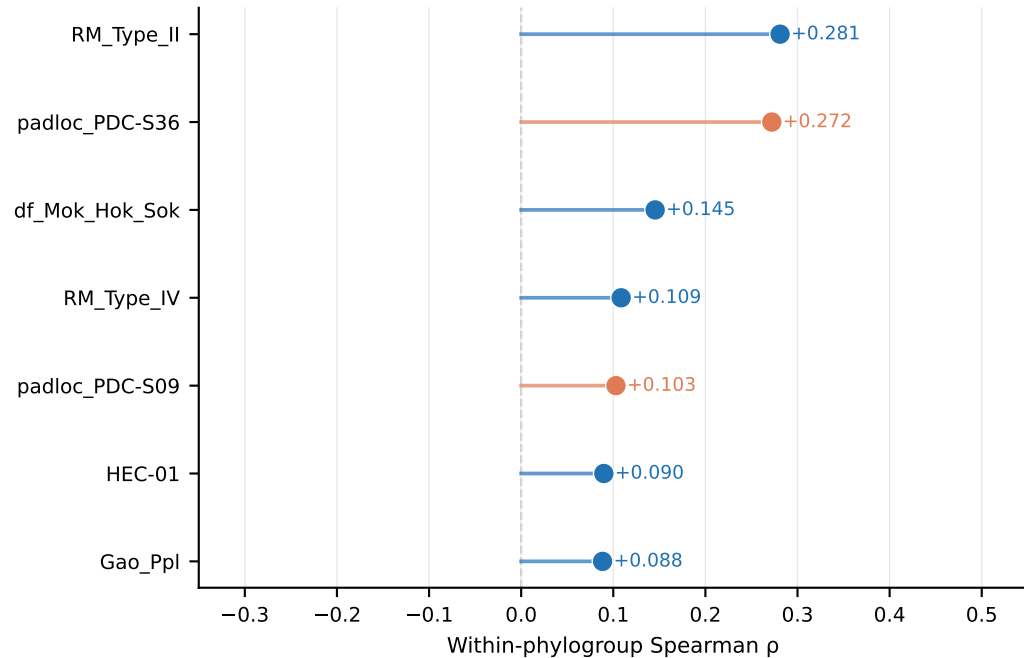
K. pneumoniae
AUROC = 0.791 (16 robust drivers; showing top 8)



P. aeruginosa
AUROC = 0.791 (16 robust drivers; showing top 8)



E. cloacae cx
AUROC = 0.781 (7 robust drivers; n = 7)



Named system Uncharacterised candidate